

Question_4:

Code:

```
1  /*
2  Question No: 4
3  Construct a Binary Search Tree using the given values and
4  print its postorder traversal.
5  Postorder:
6  Left -> Right -> Root
7  */
8
9  #include <iostream>
10 using namespace std;
11
12 struct node {
13     int data;
14     struct node *left,*right;
15 };
16 struct node* newNode(int x) {
17     node* t=new node;
18     t->data=x;
19     t->left=t->right=NULL;
20     return t;
21 }
22
23 node* insert(node* root,int x) {
24     if(!root) return newNode(x);
25
26     if(x<root->data)
27         root->left=insert(root->left,x);
28     else
29         root->right=insert(root->right,x);
30     return root;
31 }
32
33 void postorder(struct node* root) {
34     if(root) {
35         postorder(root->left);
36         postorder(root->right);
37         cout<<root->data<<" ";
38     }
```

```
39 }
40 int main() {
41     int n,x;
42     cin>>n;
43
44     node* root=NULL;
45
46     for(int i=0;i<n;i++) {
47         cin>>x;
48         root=insert(root,x);
49     }
50     postorder(root);
51 }
```

Output:

Test Case_1:

```
7
90 70 50 75 130 110 150
50 75 70 110 150 130 90
PS C:\Users\Prabin Yadav\Doc
```

Test Case_2:

```
9
50 23 56 89 43 75 130 110 150
43 23 75 110 150 130 89 56 50
PS C:\Users\Prabin Yadav\Documents\
```